

The Role of Product Variety in Targeted Pricing

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1 Introduction

2 Related Literature

The variety-seeking and state dependence literature has been primarily focused on understanding the dynamics of consumer preferences over time. Earlier papers in this literature, like those of McAlister [1982] and Lattin [1987] are most closely linked with this paper’s interpretation of variety. They posit that a consumer’s preferences for a specific product attribute is satiated with repeated purchases. McAlister finds her model of dynamic attribute satiation outperforms models with independent choice occasions in accuracy and significance of estimates. Lattin finds that there is heterogeneity in the variety-seeking types, though he uses a tiny cross-section of his own students for the study. The satiation-based approach was then built upon by Givon [1984] who used an inertia model to show how consumers can differ by preference for brand versus preference for variety. By using the brand rather than a specific attribute of a product, Givon is not able to pick up on specific drivers of product switching. Erdem and Keane [1996] also looks at the brand choice problem, modeling decision making when consumers do not have perfect knowledge of preferences over products. The authors use a Bayesian updating framework by which consumers can learn about products via experience or advertising. Their key finding is that consumers are risk averse in consumption, leading them to stay with brands they are familiar with. This model is informative as the reverse-side of my supply-side framework. Marketing researchers, Chintagunta [1999] and Dubé [2004], greatly inform the demand side of this paper. Chintagunta [1999] shows the validity of what he terms the ”Lightning Bolt” model of brand choice. He allows for time-varying variety seeking, implying the variety-seeking type can change with time. He finds that state dependence over time and marketing variables are informative in demand estimation. Dubé [2004] progresses Chintagunta’s work, showing how consumers seek variety in quantities purchased through the lens of carbonated sodas. Dubé successfully estimates consumer preferences over bundles of products within a category, finding that consumer demographics were key in determining the joint distribution. Both of these papers are primarily concerned with brand choice rather than classifying types of consumers. Finally, Ackerberg [2003] adds to Erdem and Keane’s work, allowing for consumers to learn about product attributes via advertising. The key finding here is that switching is primarily induced by the signals put out by advertisements.

The literature on personalized pricing is mostly theoretical for physical goods. Grossman and Shapiro [1984] develops a model of advertising expenditure in which the firm is trying to inform consumers of their products’ attributes. Their key finding is that advertising provides a social function by improving consumer and product matching. They also find there are excessive levels of advertising in nearly all cases. Karle and Reisinger models optimal advertising strategies for the firm. Using a theoretical model with two firms, he finds that firms’ profits change non-monotonically with targeting ability, decreasing as targeting improves before increasing again. Moorthy and Shahrokhi Tehrani [2023] derives the optimal conditions for targeted advertising. Using a Hotelling framework, the authors find that the key driver of optimal expenditure levels is product differentiation,

claiming it as a managerial framework in the face of targeting regulation. Dubé and Misra [2023] uses a randomized control trial (RCT) to understand how price personalization impacts firm profits and consumer welfare. He finds that total consumer surplus declines with personalized pricing, 60% of consumers benefit and, under some conditions, consumer welfare may increase.

My contribution is twofold. Firstly, I am primarily concerned with attribute switching rather than brand switching. Much of the previous work focuses on brand level switching. When looking at variety in attributes, I am able to see how consumers switch both within brand as well as between brands. On the supply side, I develop a simple model which can derive optimal pricing depending on consumer switching-type. This model is then used to simulate the effect of policy around targeted advertising by obscuring the firm’s information around consumer types.

3 Data

My data comes from Nielsen’s HomeScan and MarketScan data. These two main datasets give me access to household purchase decisions as well as product characteristics. From HomeScan, I can access household characteristics, purchase histories by shopping trip, and information on purchase characteristics like if a discount was used or the total price paid. From MarketScan, I get product characteristics like price or if the product was in a featured position on the shelf. I limit the sample to Atlanta, Chicago, Denver, and San Diego in 2014. These markets were chosen since they are major cities in different parts of the country. I only include single-person households to avoid any contamination of shopping for multiple people like children or spouses. This ensures I am picking up true preferences for variety rather than buying for multiple people. These trips are encoded with unique identifiers, thus allowing me to look at per-trip, and within-trip, decisions. Agents are under observation for eight weeks and asked to record each shopping trip. Stores are restricted to be valid Nielsen stores, meaning they appear in the reatil code data. This is necessary to observe product prices.

The product space I am looking at is yogurt. Yogurt was chosen for a host of reasons. First, it is a good that is not storable indefinitely, being good for approximately one to three weeks. Yogurt also has a large menu of flavors within the category, allowing consumers to switch easily along this dimension. Similarly, yogurt has many types like sugary, Greek, Skyr, low-fat, and many more. This presents potential variation to exploit as well.

I classify flavors into three categories: berry, plain, and other. Berry and other are aggregates of many flavors. Berry is comprised of any berry flavored yogurt, including mixed flavors. Other is comprised of anything which is neither berry nor plain. These flavors are determined by flavor classifications given by Nielsen. In estimation, berry and plain will be compared to other flavors, which serve as the baseline category. I define the outside option as a trip in which yogurt was not ever purchased. Table ?? shows that the outside option is taken 80% of the time each trip. My sample has 1,269 agents who are seen for an average of about 18 trips. Of those 1,269, 715 made a yogurt purchase in 2014. If an agent was a yogurt purchaser, they purchased about 2.3 cups

of yogurt per-trip. The mean income bracket for my sample was \$35,000-\$40,000. The sample is predominately white, with 75% of agents being White, 20% being Black, and 5% being either Asian or Hispanic. Men are between 25-29 years old, on average, while women are between 40-45 years old on average.

Conditional on purchasing yogurt, 57% of households switch flavors at least once during the observation period. Approximately 50% of switching households did so using a coupon during purchase. It is impossible to know what product the coupon was used on, given multiple units are purchased each period. Yogurt buying households, on average, buy the same flavor for ten consecutive trips and switch three times.

Looking at figure ??, we can see the breakdown of switching behavior by flavor. For those who switched from berry yogurt to plain yogurt, they will stay on plain for 4.4 trips on average. Those who switch to plain tend to stay on plain longer for either case, staying an average of four periods if they switch from either flavor to plain. For those switching to other, they stay on that other flavor for 3.2 trips no matter the origin flavor. For berry, it appears that households are less stuck to it or use it as their switching flavor. That is, they break up the monotony of their primary flavor, plain or other, with a berry yogurt. This is interesting, since plain is more expensive on average. This implies the household's taste for plain yogurt outweighs their disutility of cost.

Finally, when looking at pricing across markets, we see interesting patterns. Firstly, table ?? shows that prices vary a lot between cities. In Chicago, for instance, berry is seen to have a mean price of \$0.84 per unit while plain has a mean unit price of \$1.22 per cup. This is a delta approximately twice the size of that between the two flavors in other markets. Looking at figure ??, it becomes clear that the overall mean price of yogurt cups fluctuates throughout the year. In Atlanta and Denver, it seems they move largely in lockstep, while San Diego and Chicago are relatively more stable. San Diego and Chicago also have a different pattern than Atlanta and Denver do until around the middle of the year, when they start to move together. This variability across markets is what I exploit for the instrumental variable used in control function approach of Petrin and Train [2010].

4 Model

In each period t , the consumer i chooses a product j from the menu of available products J_t . The available products are $J_t = \{0, 1, 2, 3\}$ where $j = 0$ is the outside option, $j = 1$ is purchasing berry yogurt, $j = 2$ is purchasing plain yogurt, and $j = 3$ is purchasing other flavors. Consumers are assumed to be state-dependent, carrying previous choices forward when love of variety is present. Otherwise, they only internalize characteristic utility and price utility. Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X_{jt}^k \in \mathcal{X}_{jt}$. The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \beta_0 + \beta_{it}X_{jt} + \gamma_i\Xi_{ijt} + \alpha_ip_{jt} + \varepsilon_{ijt}, \text{ for } j \in \{1, 2, 3\}$$

such that

$$\begin{aligned}\Xi_{ijt} &= \lambda_i \sum_{s=1}^8 \delta^{t-s} \Xi_{ijs} + (1 - \lambda_i) \sum_{j \neq k} \rho x_{k,t-1} \\ \varepsilon_{ijt} &\sim \text{T1EV}(0, 1) \\ U_{i0t} &= \varepsilon_0\end{aligned}$$

In the utility function, there are the following components: β_0 is a constant, β_{it} is a consumer's utility for characteristic X_{jt} associated with product j in time t . Following Lattin [1987], love of variety preference is captured by γ_i . A consumer's history is given by Ξ_{ijt} where λ_i is a weight on state dependence Ξ_{ijs} which is discounted with time δ^{t-s} . ρ is an indicator for if a consumer bought a product $x_{k,t-1}$ where $k \neq j$. α_i is the parameter on prices p_{jt} and ε_{ijt} is the Gumbel shock.

Choice probabilities take the following form:

$$P_{ijt} = \frac{\exp(\beta_0 + \beta_{it}X_{jt} + \gamma_i\Xi_{ijt} + \alpha_ip_{jt} + \varepsilon_{ijt})}{1 + \sum_k \exp(\beta_0 + \beta_{it}X_{kt} + \gamma_i\Xi_{ikt} + \alpha_ip_{kt} + \varepsilon_{ikt})}$$

Where we normalize the outside option, not purchasing yogurt, to be zero by construction.

5 Estimation

Following Petrin and Train [2010], I use a control function to control for price endogeneity. I do a first stage instrumental variable estimation using Hausman instruments and then take residual from that estimation and insert it into the utility function. The first stage takes the form:

$$p_{jt} = \bar{p}_{j,-m(i),t} + \tau_t + \eta_{jt}$$

Where $\bar{p}_{j,-m(i),t}$ is the mean price from the three markets the household is not shopping in and τ_t is a time indicator. The mean price of the other three markets was chosen as an instrument to separate any region specific effects like local advertising. Thus, I assume there is no national advertising or other effects which impact demand across all markets. The residual recovered from the first stage corrects for the bias present in pricing, allowing us a consistent estimate of pricing.

I make an index of prices, instrumental variable residuals, and previous choices for each household's trips for each flavor. These are then used to estimate the objective function:

$$U_{ijt}^k = \beta_0 + \sum_k \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha p_{jt} + \sigma \hat{\eta}_{jt}$$

Where β_0 is a constant, β^k for $k \in \{\text{berry, other, plain}\}$ are choice specific coefficients for choosing berry, other, or plain. γ is the love of variety term, interacted with Ξ_{ijt} , which is just capturing if consumer i switched between periods. The price and control function term are unchanged from above. The probability of individual i choosing the observed product j in time t is given by:

$$\Pi_i(P_{ijt}(\beta_i, \Xi_{ijt}))^{y_{ijt}}$$

Where $y_{ijt} = 1$ when person i chosen j that period, and zero otherwise. The log-likelihood is then given by:

$$\text{LL}(\theta) = \sum_t \sum_i \sum_j y_{ijt} \ln P_{ijt}(\beta_i, \Xi_{ijt})$$

I also repeat estimation under specifications which exclude γ_i , do not estimate heterogeneous β_i , γ_i , and do both. This allows me to see if results are biased when love of variety is excluded from estimation.

6 Supply Side Model

There are N consumers, each endowed with a type γ_r for $r \in \{H, L\}$, corresponding to high or low utility from variety in products consumed. As such, there are $N_r \in N$ consumers of each type, $r \in \{H, L\}$. Consumers maximize utility according to the following:

$$U_{irjt} = \beta_0 + \gamma_r \{j_t \neq j_{t-1}\} + \alpha p_{rjt} + \varepsilon_{irjt}$$

Where utility is over individuals $i \in N$, type $r \in \{H, L\}$, products $j \in J$, and time $t \in T$. β_0 is an intercept, representing the baseline utility from consuming. γ_i is the consumer's utility from consuming different products, and is active if they switched products from one period to the next. α is disutility of price, which can be lowered via a discount from advertising. Finally, ε_{ijt} is the Gumbel error, giving us a closed form Logit.

This equation can be written in terms of deterministic vs. random utility:

$$\begin{aligned}
U_{irjt} &= V_{irjt} + \varepsilon_{irjt} \\
&\text{s.t.} \\
V_{irjt} &= \beta_0 + \gamma_r \{j_t \neq j_{t-1}\} + \alpha(p_{jt} - d_{rjt})
\end{aligned}$$

Choice probabilities of consuming the inside option(s), as well as the outside option utility normalized to be zero, is written as such:

$$\begin{aligned}
P_{irjt} &= \frac{e^{V_{irjt}}}{1 + \sum_k e^{V_{irk t}}} \\
P_{ir0t} &= \frac{1}{1 + \sum_k e^{V_{irk t}}}
\end{aligned}$$

Next, because consumers of the same type r are homogenous, we can say:

$$s_{rjt} = \sum_i P_{irjt} \text{ for } r \in \{H, L\}$$

Taking the derivative with respect to prices, we can get the own and cross price elasticities:

$$\begin{aligned}
\epsilon_{jj}^p &= \alpha p_{jt} (1 - s_{rjt}) \\
\epsilon_{jk}^p &= -\alpha p_{kt} s_{rkt} \text{ for } k \neq j
\end{aligned}$$

Thus, since $\alpha < 0$, we can see that own price elasticity is negative and cross-price is positive.

Now, a supermarket enters the market. The supermarket is able to control prices $\bar{p}_j$ for a product j . The supermarket can also offer a discount d_{rjt} which is tailored to our switching types $r \in \{H, L\}$. The firm's actual price is written as:

$$\begin{aligned}
p_{rjt} &\equiv \bar{p}_j - d_{rjt} \\
&\text{s.t.} \\
d_{rjt} &> 0
\end{aligned}$$

So, as the market raises the discount, utility increases as $\alpha < 0$ so $-\alpha d_{rjt} > 0$. It is through this discounting that firm can induce switching. The firm's profit margin for a product j is given by

$$\begin{aligned}
m_{rjt} &= p_{rjt} - c_j \\
&= \bar{p}_j - d_{rjt} - c_j
\end{aligned}$$

This says that as we raise the discount, we raise market share on the product being discounted but lose profit margin.

With these definitions in mind, we will be looking at a two period model under two specifications. First, the case where the firm has perfect information on the consumers' types and can thus perfectly price. In the second case, we will look at the case of imperfect information, where the firm is using Bayesian updating to approximate the type distribution.

6.1 Perfect Information

Assume the firm has perfect information over the distribution of types. That is, the supermarket perfectly observes each consumer type r and thus knows the mass N_r of each type. Define period one profit, π_1 as such:

$$\begin{aligned}
\pi_1 &= \sum_r N_r \sum_j s_{rj1}(m_{rj1}) \\
&\quad \text{s.t.} \\
s_{rj1} &= P_{irj1} \\
m_{rj1} &= \bar{p}_j - d_{rj1} - c_j
\end{aligned}$$

Here, the firm is understanding how each type's share for each product impacts their profits. We can similarly define period two profits, π_2 like so:

$$\pi_2 = \sum_r N_r \sum_j s_{rj1} \left[\sum_k s_{rk2}(j)(m_{rk2}(j)) \right]$$

The only augmentation is the firm now understands all of the possible paths for a consumer of type r who chose product j in period 1, and product k in period 2. For the sake of conciseness in notation, $w_r(j) = \sum_k s_{rk2}(j)(m_{rk2}(j))$. Define total profit, Π , as such:

$$\begin{aligned}
\Pi &= \pi_1 + \pi_2 \\
&= \sum_r N_r \sum_j s_{rj1}(m_{rj1} + \delta w_r(j))
\end{aligned}$$

Where δ is a discount factor on future profits. Taking the derivative with respect to period one

prices for a product j yields:

$$\frac{\partial \Pi}{\partial p_{rj1}} : 0 = 1 + \alpha \sum_{k \neq j} s_{rk1} [(m_{rj1} - m_{rk1}) + \delta(w_r(j) - w_r(k))]$$

The firm sets prices according to the current difference in profit margins and the difference in future profits which occur upon switching a consumer between product j and product k . When we do the same for period two, we recover the following:

$$\frac{\partial \Pi}{\partial p_{rk2}} : 0 = 1 + \alpha \sum_{l \neq k} s_{rl2}(j) [(m_{rk2}(j) - m_{rl2}(j))]$$

Now, expanding the expression and solving for period one prices:

$$\begin{aligned} 0 &= 1 + \alpha \left[\sum_{k \neq j} s_{rk1} ((p_{rj1} - c_j) - (p_{rk1} - c_k) + \delta(w_r(j) - w_r(k))) \right] \\ 0 &= 1 + \alpha(p_{rj1} - c_j) \sum_{k \neq j} s_{rk1} - \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) + \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ \alpha(p_{rj1} - c_j) \sum_{k \neq j} s_{rk1} &= -1 + \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) - \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ \alpha p_{rj1} \sum_{k \neq j} s_{rk1} &= \alpha c_j \sum_{k \neq j} s_{rk1} - 1 + \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) - \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ p_{rj1}^* &= c_j - \frac{1}{\alpha \sum_{k \neq j} s_{rk1}} + \frac{\sum_{k \neq j} s_{rk1} (p_{rk1} - c_k)}{\sum_{k \neq j} s_{rk1}} - \delta \frac{\sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k))}{\sum_{k \neq j} s_{rk1}} \\ p_{rj1}^* &= c_j - \frac{1}{\alpha(1 - s_{rj1})} + \frac{\sum_{k \neq j} s_{rk1} m_{rk1}}{1 - s_{rj1}} - \delta \frac{\sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k))}{1 - s_{rj1}} \end{aligned}$$

From this, the firm with perfect information sets each type r 's price to equal the cost plus a demand markup and the share weighted average margin of alternatives $k \neq j$ minus the intertemporal tradeoff. Here, the firm will be willing to lower the price of good j today if it implies higher profits tomorrow ($w_r(j) > w_r(k)$), increasing probability of purchase. In the opposite case, the firm would raise prices on product j to encourage shifting from product j to product k tomorrow. That is, they use the past choice to determine consumer state and thus future profits for each product. The firm again thinks of the tradeoff, without having to think about future profit as period two is terminal. Under perfect information, the supermarket chooses its prices by evaluating the current margin and future profit associated with reallocating consumers across products. Now, we move to the case of imperfect information.

6.2 Imperfect Information

In this case, the supermarket does not know consumer types with certainty, nor do they know the mass of either type. So, they must infer the distribution from consumer purchases. Define the following objects:

$$\begin{aligned}\lambda_1 &= Pr(r = H) \\ \lambda_2 &= Pr(r = H|j) \Rightarrow \lambda_2(j) = \frac{\lambda_1 s_{Hj1}}{\lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1}} \\ \tilde{s}_{j1} &= \lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1} \\ \tilde{s}_{k2}(j) &= \lambda_2(j) s_{Hk2}(j) + (1 - \lambda_2(j)) s_{Lk2}(j)\end{aligned}$$

Now, we can define the profit functions much like we did before:

$$\begin{aligned}\pi_1 &= N \sum_j \tilde{s}_{j1} m_{j1} \\ \pi_2 &= N \sum_j \tilde{s}_{j1} \sum_k \tilde{s}_{k2}(j) m_{k2}(j)\end{aligned}$$

The key difference here is that the firm must now update their beliefs from period to period rather than being able to discount according to the tradeoffs between two products. This implies that there is room for error in discounting. When combining the two into overall profit, we get:

$$\Pi = N \sum_j \tilde{s}_{j1} [m_{j1} + \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j)]$$

Now, we can take the derivative with respect to period one prices for product j :

$$\begin{aligned}\frac{\partial \Pi}{\partial p_{j1}} : 0 &= N \tilde{s}_{j1} + \alpha N [\lambda_1 s_{Hj1} (1 - s_{Hj1}) + (1 - \lambda_1) s_{Lj1} (1 - s_{Lj1})] [m_{j1} + \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j)] \\ &\quad - \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1 - \lambda_1) s_{Lh1} s_{Lj1}] [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\ &\quad + \delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} [\pi_{H2}(h) - \pi_{L2}(h)]\end{aligned}$$

This is much the same as the perfect information case. However, looking at the third line, we can see where the firm incorporates their belief updating. Define the derivative in the third line as such:

$$\frac{\partial \lambda_2(h)}{\partial p_{j1}} = \begin{cases} 0 = \frac{\alpha \lambda_1 (1 - \lambda_1)}{\tilde{s}_{j1}^2} [s_{Hj1} s_{Lj1} (s_{Lj1} - s_{Hj1})] & \text{if } h = j \\ 0 = \frac{\alpha \lambda_1 (1 - \lambda_1)}{\tilde{s}_{h1}^2} [s_{Hh1} s_{Lh1} (s_{Lj1} - s_{Hj1})] & \text{if } h \neq j \end{cases}$$

From this, the firm updates its belief about the distribution of consumer types upon observing period one consumption. The magnitude and direction of this update depends on how likely that choice is for each type. In total, the third line of the derivative interacts the probability of purchasing item h with the effect of price on the firm's posterior belief and the value of its belief in one type or the other.

Now, we can solve for p_{j1} via the following steps. First,

$$\begin{aligned} \alpha N \mathbb{A} [p_{j1} - c_j + \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j)] = \\ - N \tilde{s}_{j1} + \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1 - \lambda_1) s_{Lh1} s_{Lj1}] [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\ - N \delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} [\pi_{H2}(h) - \pi_{L2}(h)] \end{aligned}$$

Now, we isolate p_{j1} :

$$\begin{aligned} \alpha N \mathbb{A} p_{j1} = \alpha N \mathbb{A} c_j - \alpha N \mathbb{A} \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j) \\ - N \tilde{s}_{j1} + \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1 - \lambda_1) s_{Lh1} s_{Lj1}] [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\ - N \delta \sum_h \tilde{s}_{h1} \left[\frac{\partial \lambda_2(h)}{\partial p_{j1}} (\pi_{2H}(h) - \pi_{2L}(h)) \right] \end{aligned}$$

And simplify:

$$\begin{aligned} p_{j1} = c_j - \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j) - \frac{\tilde{s}_{j1}}{\alpha \mathbb{A}} \\ + \frac{\sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1 - \lambda_1) s_{Lh1} s_{Lj1}]}{\mathbb{A}} [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\ - \frac{\delta}{\alpha \mathbb{A}} \left[\sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} (\pi_{2H}(h) - \pi_{2L}(h)) \right] \end{aligned}$$

Deconstructing the optimal price, optimal price is composed of the marginal cost, minus the own-

product discounted continuation value, plus a demand markup term and the the value of alternative products, and lastly a term referring to the learning adjustment from period one behavior. In the imperfect-information specification, today’s prices affect current demand and the distribution of future states as well as the information that can be learned from consumer purchases. Period one prices alter the likelihood of purchase, thus affecting the posterior over consumer type and, by extension, optimal pricing in period two.

7 Results

Results are shown in table ?? . I find that γ_i is statistically significant and negative with a value of -1.23 . This finding points to consumers being less willing to switch flavors between trips. This estimate is at odds with the descriptive facts: more than half of houses are switchers, conditional on yogurt purchase, and houses switch three times on average if they do switch. This is likely a byproduct of not having random coefficients for utility from a given product, instead taking an average across households. Further, I find a negative price coefficient α , but am unable to claim statistical significance. This is fixable with a better instrumental variable approach, something which is actively being considered. Similarly, the control function correction coefficient is not significant either. However, given σ is positive, we can infer that α is positively biased without this correction. With a better IV approach, the control function should also produce more impactful effects. I am looking into factors like distribution networks, other product categories, as well as different sources of geographic variation to serve as instruments.

When looking at the characteristic coefficients, only berry, β_i^b , is statistically significant. This is, again, an issue with aggregate level coefficients. When looking at plain’s coefficient, β_i^p , I do not find significance. The constant is negative, which is to be expected with a high outside option uptake rate. It is close to fringe significance, but is not either.

Finally, I calculated a willingness-to-pay for all β coefficients as well as the variety coefficient γ . These values are found in table ?? . For the intercept, consumers value the outside option approximately \$1.50 more than yogurt flavored "other". For berry, it is valued about \$1.81 more than other flavors, while plain is valued about \$0.77 more. More interestingly, I find that consumers would need to be heavily compensated to switch. They would need approximately \$2.20 in incentives to switch from their normal choice of flavor. Judging by the average pricing of yogurt, this implies needing about two to three cups of yogurt given to them for free to switch.

Overall, my findings suggest consumers prefer to not switch flavors. While a surprising result given the data facts, it must be noted that consumers do not just buy flavor. They buy for other characteristics that are not accounted for at this point in the research process. It will be interesting to see if this changes once I create a more holistic bundle of characteristics for the consumer to have preferences over. Similarly, the willingness-to-pay for variety is -2.20 , meaning the consumer would need to be compensated \$2.20 to switch flavors. This opens a door for interesting counterfactual exercises in which the supply side is aware of the consumers willingness-to-pay and preferences, and

thus try to influence them to switch.

8 Counterfactuals

Now, suppose firm side dynamics are introduced. Here, the firm acts strategically in a competitive market, taking the consumer’s last observed action into account, and deciding whether or not to offer a promotion to encourage switching. This would lead to a dynamic pricing problem in which the firm is weighing the potential markup in the future against a lower price today. The firm will be able to observe consumers’ purchase histories, enabling them to first degree price discriminate. A firm would choose to discount a product j if the future market share it obtains offsets the lowered profits today. This also leads to analysis of competitive behavior between firms and how they decide to price depending on the behavior of other suppliers. Because the firm would not directly observe each consumer’s type, I would be able to see the impact of targeting error on three dimensions: consumer welfare, competition in the market, and firm profits.

I also aim to look at a counterfactual involving product introduction. The firm would take its present approximation of the distribution over preferences and decide whether or not to introduce a product. Using a Hotelling model, I can test the impact of product introduction on switching behavior. With this, I am interested in consumer welfare, optimal product differentiation, and competitive effects.

9 Conclusion

In closing, this paper implements a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic pricing by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on love of variety, price-discrimination, and dynamic consumer behavior, by providing empirical evidence on the effects of strategic pricing in a dynamic demand setting.

I find that consumers experience disutility from switching, needing to be compensated \$2.20 to switch from one flavor to another. I find insignificant pricing effects, even with endogeneity intervention implying a need for a stronger instrumental variable. This simple version of the model suggests that there is significant runway for saturation with other product characteristics, random coefficients, and a richer sample.

The next iteration of this work will first augment the sample to include 2022-2024, as well as add markets. Next, I will nail down the function form of utility to better implement the idea of attribute satiation. There is also room to add more characteristics like calories, grams of protein, grams of sugar, and type of yogurt. A brand effects pathway also makes sense, seeing if households are brand loyal or loyal to flavors. This decomposition of switching behavior would be very useful for the product introduction counterfactual especially. Lastly, the supply side will be an ongoing

effort. With that being my primary contribution, it is obviously a priority and thus will be a major focus of forthcoming research.

References

- Daniel A. Akerberg. Advertising, learning, and consumer choice in experience good markets: an empirical examination*. *International Economic Review*, 44(3):1007–1040, August 2003. ISSN 0020-6598, 1468-2354. doi: 10.1111/1468-2354.t01-2-00098. URL <https://onlinelibrary.wiley.com/doi/10.1111/1468-2354.t01-2-00098>.
- Pradeep K. Chintagunta. Variety Seeking, Purchase Timing, and the “Lightning Bolt” Brand Choice Model. *Management Science*, 45(4):486–498, April 1999. ISSN 0025-1909, 1526-5501. doi: 10.1287/mnsc.45.4.486. URL <https://pubsonline.informs.org/doi/10.1287/mnsc.45.4.486>.
- Jean-Pierre Dubé and Sanjog Misra. Personalized pricing and consumer welfare. *Journal of Political Economy*, 131(1):131–189, None 2023. doi: 10.1086/720793. URL <https://ideas.repec.org/a/ucp/jpolec/doi10.1086-720793.html>.
- Jean-Pierre Dubé. Multiple Discreteness and Product Differentiation: Demand for Carbonated Soft Drinks. *Marketing Science*, 23(1):66–81, February 2004. ISSN 0732-2399, 1526-548X. doi: 10.1287/mksc.1030.0041. URL <https://pubsonline.informs.org/doi/10.1287/mksc.1030.0041>.
- Tülin Erdem and Michael P. Keane. Decision-Making Under Uncertainty: Capturing Dynamic Brand Choice Processes in Turbulent Consumer Goods Markets. *Marketing Science*, 15(1): 1–20, February 1996. ISSN 0732-2399, 1526-548X. doi: 10.1287/mksc.15.1.1. URL <https://pubsonline.informs.org/doi/10.1287/mksc.15.1.1>.
- Moshe Givon. Variety Seeking Through Brand Switching. *Marketing Science*, 3(1):1–22, February 1984. ISSN 0732-2399, 1526-548X. doi: 10.1287/mksc.3.1.1. URL <https://pubsonline.informs.org/doi/10.1287/mksc.3.1.1>.
- Gene M. Grossman and Carl Shapiro. Informative Advertising with Differentiated Products. *The Review of Economic Studies*, 51(1):63, January 1984. ISSN 00346527. doi: 10.2307/2297705. URL <https://academic.oup.com/restud/article-lookup/doi/10.2307/2297705>.
- Heiko Karle and Markus Reisinger. Imperfect Targeting and Advertising Strategies.
- James M. Lattin. A Model of Balanced Choice Behavior. *Marketing Science*, 6(1):48–65, February 1987. ISSN 0732-2399, 1526-548X. doi: 10.1287/mksc.6.1.48. URL <https://pubsonline.informs.org/doi/10.1287/mksc.6.1.48>.
- Leigh McAlister. A dynamic attribute satiation model of variety-seeking behavior. *Journal of Consumer Research*, 9(2):141–50, 1982. URL <https://EconPapers.repec.org/RePEc:oup:jcnrs:v:9:y:1982:i:2:p:141-50>.
- Sridhar Moorthy and Shervin Shahrokhi Tehrani. Targeting Advertising Spending and Price on the Hotelling Line. *Marketing Science*, 42(6):1057–1079, November 2023. ISSN 0732-2399, 1526-548X.

doi: 10.1287/mksc.2022.1422. URL <https://pubsonline.informs.org/doi/10.1287/mksc.2022.1422>.

Amil Petrin and Kenneth Train. A Control Function Approach to Endogeneity in Consumer Choice Models. *Journal of Marketing Research*, 47(1):3–13, February 2010. ISSN 0022-2437, 1547-7193. doi: 10.1509/jmkr.47.1.3. URL <https://journals.sagepub.com/doi/10.1509/jmkr.47.1.3>.

10 Tables and Figures